

Screening LCA and TEA of Municipal Solid Waste Recovery Pathways in Indonesia

Methodology v3.1: 2025 tonnage baseline, Level I/II/III taxonomy, auditable material parameters, managed-waste candidates, fossil energy and land use, decision rules from scenario discovery, and a reproducible decision tool

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Supervisor: Prof. Anthony Halog — October 2026 — Companion code: `msw_pathway_model_v3.py` / `MSW_pathway_model_v3.ipynb`; user tool: `MSW_Decision_Tool_Colab.ipynb`

What changed in v3 and what it shows

- **One source of truth.** Every parameter, constant, coordinate and threshold is read from `data/*.csv`, which `build_database_v3.py` builds from the update package. The model file holds no numbers. 60 of 60 validation checks pass (Section 7).
- **2025 baseline with three separate years.** Composition sampling year, tonnage source year and baseline year are stored separately. Tonnage is projected once, $Q_{2025} = Q_t(1+g)^{2025-t}$; the parser refuses the old file, so a second projection cannot happen. 8 overview totals have an unverified year; they are used as reported and a projected alternative is run as a scenario. They are not labelled as 2025 observations.
- **Composition is a proxy.** RIPS percentages from the sampling year (2014 to 2025) are normalised per stream (factor recorded) and used for 2025 without any trend. They are not a measured 2025 composition.
- **Managed waste (Table 5) kept in two series.** M1 is the RIPS status index (19 locations); M2 uses a local indicator where one is held. The two use different definitions; differences reach 86 percentage points. Padang's 93.71% is provisional (the 141.76 t/d recovery may be a potential); Serang's 14% is a target, not a baseline.
- **Taxonomy and parameters.** 10 Level I groups, 36 Level II fractions and a provisional catalogue of 56 Level III refinements, keyed by (level, code). Level II/III parameters are parent-category proxies or explicit gaps; Level II/III allocations are priors and never enter the model results. Rubber/leather DOCf is a declared gap (scenario 0 to 0.5).
- **Results barely move; their status is clearer.** At central values sanitary landfill with flare has the lowest social cost everywhere up to USD 25/t CO₂e; RDF + AD is best in 18 locations at USD 50 and 20 at USD 100. Over the sampled carbon range, landfill is most probable in 17 locations and RDF + AD in 4, but 6 of the 21 leads are below 0.05. The ranking is sensitive to the moisture assumption: with IPCC default moisture RDF + AD becomes most probable in 20 locations.
- **New in v3.1: conditions, energy, land, tool.** Scenario discovery over 40,000 combinations of city conditions turns the model into decision rules (Section 11). Fossil cumulative energy demand and land take are added (Section 10). Sobol indices confirm that as-received moisture and landfill-gas collection decide the landfill versus RDF + AD comparison. A Colab notebook lets other researchers and planners run the model for their own cities (Section 12).
- **Still provisional.** Composition proxies of different years, legacy heating values and RDF transfer coefficients, status L literature values and class-5 costs. Section 13 states the limits of validity.

1 Scope of the update and vocabulary of evidence

This document updates the v2 methodology with the research package of October 2026 (folders 02 to 04). The package was read in the order its START_HERE file prescribes: the v2 code and PDF, then the assumption audit (ASSUMPTION_SOURCE_AUDIT.md), which overrides the interpretation of the older allocation workbook, then the 2025 baseline files, then the fraction-literature files. The SHA-256 of every source file was checked against the package manifest before use. The equations, system boundary and decision rule of v2 are kept; what changes is where numbers come from, how their status is recorded and how the 2025 baseline is formed.

Every value in the database carries one of the following statuses. A blank cell always means *unknown or not applicable*; it is never read as zero.

Status	Meaning	Example
local measurement	Reported for the location by its RIPS document	RIPS wet-mass composition, Kab. Bogor 2022
literature default	Published default for a category, not for the location	IPCC 2006 Table 2.4 DOC of food, 0.38 (dry)
parent-category proxy	A parent default assigned to a sub-fraction; not a sub-fraction measurement	Food DOC for 1.2.3 unavoidable animal-derived food
foreign benchmark	Measured elsewhere on a different waste stream	Edjabou et al. (2015) Danish residual household waste
assumption / prior	Analyst choice, to be defended or replaced	Organic split 85/15 (A01); moisture as received
scenario	Alternative value run to show sensitivity, never a baseline	Serang 14% target; rubber DOCf 0.5

Status	Meaning	Example
gap	No defensible number found; left blank, scenario if the model needs one	Rubber/leather DOCf; WEEE material composition

Table 1: Vocabulary of evidence used in every CSV and table of this document.

2 Goal, functional unit and system boundary (unchanged)

The research question remains: *under what Indonesian city and waste-system conditions does each recovery pathway become preferable?* The functional unit is the management of 1 tonne of mixed MSW, wet weight, as received at the gate of the treatment facility, with the composition of the location, in the reference year 2025. Six options are compared within one LCA/TEA boundary, from the facility gate to final disposal of residues and delivery of products, with system expansion for displaced electricity, coal and polypropylene: sanitary landfill with flare (SL), WtE (S1), RDF to cement kiln (S2), AD of food waste (S3), PHB from landfill gas (S4) and integrated RDF + AD (S5). Two baselines are reported: open dump (OD) and sanitary landfill (SL). Climate change is assessed with GWP100 of IPCC AR6 (GWP20 as a sensitivity case); cost is a net levelised cost in USD per tonne (AACE class 5). These choices and their justification are those of v2, Sections 2 and 3.

3 Data harmonisation for the 2025 baseline

3.1 Three years, kept apart

Each location now carries a composition sampling year t_c , a tonnage source year t (blank when it is unverified) and the baseline year 2025. Tonnage is harmonised with

$$Q_{2025} = Q_t (1 + g)^{2025 - t} \quad M_{2025}^d = M_t^d (1 + g)^{2025 - t}, \quad M_{2025}^n = M_t^n (1 + g)^{2025 - t} \quad (1)$$

with g the RIPS growth rate where one is given (6 locations) and otherwise the median of those six rates, 1.73%/yr (assumption H11, range 0.65 to 2.54%). Domestic and non-domestic streams use the same factor, so $M_{2025}^d + M_{2025}^n = Q_{2025}$. The exponent is zero when the tonnage already refers to 2025. The flag *tonnage_already_2025* is set in the CSV and read by the parser, which uses Q_{2025} directly; an assertion in the model stops the run if the tonnage it computes differs from the CSV baseline (single-projection guard).

Class	n	Rule (central)	Locations
stated in source table	12	projected once from the stated year	Kota Padang, Kab. Bogor, Kab. Serang, Kota Denpasar, Kab. Cianjur, Kab. Indramayu, Kab. Pati, Kab. Pandeglang, Kab. Temanggung, Kab. Pekalongan, Kota Pekalongan, Kota Magelang
overview index year unverified	8	used as reported; alternative projected from t_c	Kota Semarang, Kab. Brebes, Kab. Nganjuk, Kab. Magelang, Kab. Sragen, Kab. Kutai Kartanegara, Kab. Kendal, Kab. Wonogiri
RIPS projection for 2025	1	used as reported; flagged as a projection	Kab. Pemalang

Table S1: Tonnage-year classes. The overview class also covers Kota Semarang, Kab. Sragen and Kab. Kutai Kartanegara, whose source pages cite the same overview index although v2 and the package projected them from the composition year.

Decision on overview totals. The package treated five overview totals as 2025 values on the basis of a keyword in free-text notes, while three other rows citing the same overview index were projected. Neither the year of the overview index nor its relation to the composition year is shown in the rows held. v3 therefore applies one rule to all eight: the total is used as reported (no growth applied, so no double projection if it is already current) and its year is left blank with flag T1. The alternative, that the total dates from the composition year, is computed for every row ($Q_{2025_alt_tpd}$) and run as the scenario *tonnage_overview_projected*. It raises tonnage by up to 21% (Kab. Kendal, composition 2014) and changes no most-probable pathway (Section 8). Kab. Pemalang's 2025 tonnage is itself a RIPS projection (population $\times$ 0.31 kg/cap/day) and is flagged T2.

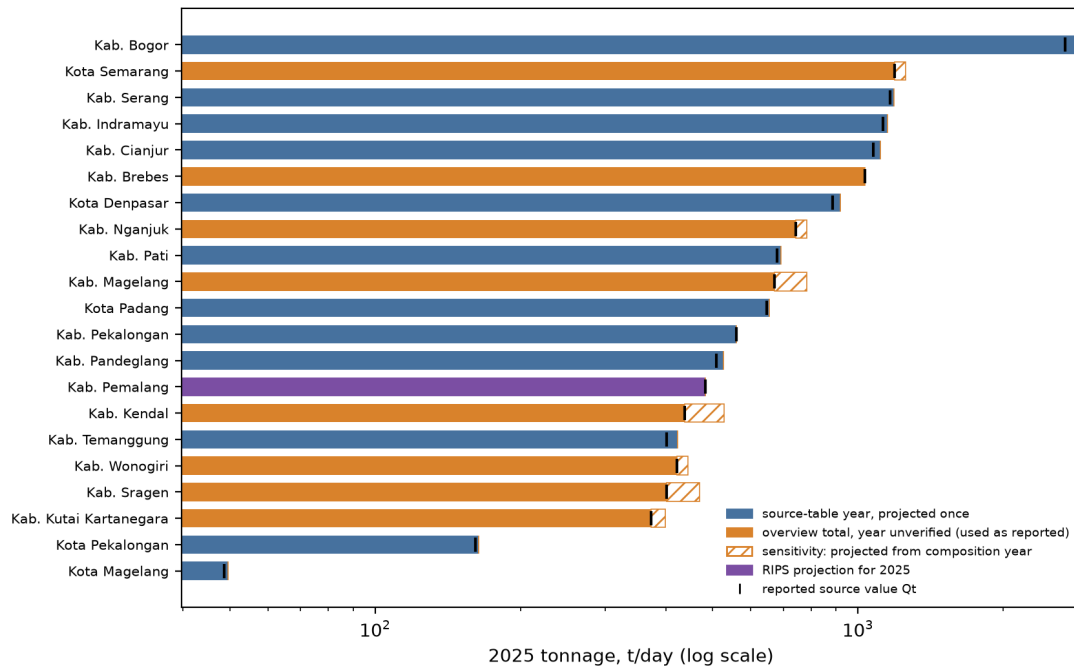


Figure 1: 2025 tonnage by location and tonnage-year class. Hatched: the projected alternative for overview totals. Ticks: reported source value Q_i .

3.2 Composition: a sampling-year proxy, normalised per stream

RIPS compositions are wet-mass percentages measured in the sampling year with the SNI 19-3964-1994 method. No source gives a defensible annual composition trend, so the shares are used for 2025 unchanged and labelled *RIPS sampling-year composition held constant as 2025 proxy* (flag C0). Growth is never applied to percentages. Rounded source totals (99.99 to 100.10%) are normalised per stream k and the factor is stored:

$$S_j^{(k)} = \frac{x_j^{(k)}}{\sum_i x_i^{(k)}}, \quad f^{(k)} = \frac{100}{\sum_i x_i^{(k)}}, \quad S_j = \frac{M_{2025}^d S_j^{(d)} + M_{2025}^n S_j^{(n)}}{M_{2025}^d + M_{2025}^n} \quad (2)$$

Kab. Pemalang (100.10%) has $f = 0.9990$. Blank category cells are absent categories: glass is *reported* in 15 locations, *folded into lain-lain* in 5 and *not reported* in Kota Padang, and the three cases are flagged separately (C3, C4). Rules L (organic aggregate split with share γ) and W (kayu treated as garden) of v2 are kept as sampled harmonisation priors. Semarang and Nganjuk report a non-domestic composition but no non-domestic tonnage; it receives zero weight (flag C6). All compositions refer to waste at generation or collection, not at the landfill gate.

3.3 Managed waste and Table 5

The managed share μ is used only in the managed-tonnage scenarios, where it scales the plant size: $Q_{\text{managed}} = \mu Q_{2025}$. Holding μ constant from its source year is a scenario assumption, not a 2025 observation. The package filled the seven blanks of v2 with candidates whose definitions differ; v3 keeps every candidate visible instead of choosing by numerical similarity:

Indicator family	Locations	Definition and status
RIPS status index (M1)	14 legacy + 5 candidates (Bogor, Denpasar, Cianjur, Indramayu, Semarang)	Same index series as v2 Table 5. Its definition and denominator are not documented in the package rows; comparable only within the series.
Local service / handling indicators	Bogor 35.15%, Denpasar 96.91%, Cianjur 30.36%, Indramayu 61.52%, Semarang 59.43%	Urban managed waste, reduction + handling, or service level. Different numerators and denominators from the index; kept as alternatives, not merged.
Padang, derived	93.71% (upper), 71.82% (TPA delivery only)	(141.76 recovery + 465.08 to TPA) / 647.57. Source is a draft feasibility study (p. 94); the recovery may be a potential and the denominator is the 2023 RIPS total. Provisional; not an observed managed share.
Serang, local observed	7.45% (14% target as scenario only)	Existing condition from the master-plan executive summary (p. 154); the 14% is a 2025 service target (pp. 180-181).

Table S2: Families of managed-waste indicators held in the database.

Two series are run and reported side by side. **M1** uses the status index only: 19 locations, Padang and Serang n.a. **M2** uses the local indicator where one is held (Padang at its 71.82% lower bound, because the recovery numerator is

unverified) and the status index elsewhere; it mixes definitions by construction and is labelled so. Neither series implies that all 21 locations have the same indicator. $Q_{\text{managed}} \leq Q_{2025}$ holds for every value.

Location	Q2025 t/d	Status index	Local	M1 t/d	M2 t/d	Status of the local value
Kota Padang	656	n.a.	93.71% (71.82% low)	n.a.	471	derived, provisional (not an observed managed share)
Kab. Bogor	2,891	22.10%	35.15%	639	1016	local indicator with a different definition from the status index
Kab. Serang	1,187	n.a.	7.45%	n.a.	88	local observed (existing condition)
Kota Denpasar	918	10.73%	96.91%	99	890	local indicator with a different definition from the status index
Kab. Cianjur	1,114	2.67%	30.36%	30	338	local indicator with a different definition from the status index
Kab. Indramayu	1,152	8.59%	61.52%	99	709	local indicator with a different definition from the status index
Kota Semarang	1,190	8.27%	59.43%	98	707	local indicator with a different definition from the status index
Kab. Brebes	1,033	2.32%	-	24	24	
Kab. Nganjuk	744	11.33%	-	84	84	
Kab. Pati	691	18.38%	-	127	127	
Kab. Magelang	671	10.45%	-	70	70	
Kab. Pandeglang	527	29.80%	-	157	157	
Kab. Pemalang	484	16.11%	-	78	78	
Kab. Temanggung	423	8.65%	-	37	37	
Kab. Pekalongan	560	15.58%	-	87	87	
Kab. Sragen	402	20.24%	-	81	81	
Kab. Kutai Kartanegara	373	55.92%	-	209	209	
Kab. Kendal	437	27.41%	-	120	120	
Kab. Wonogiri	422	20.02%	-	84	84	
Kota Pekalongan	164	48.16%	-	79	79	
Kota Magelang	50	68.49%	-	34	34	

Table 5: Managed waste in 2025 (constant-share scenario). Status-index values of the 14 original locations are carried over from v2 without re-audit of their definition.

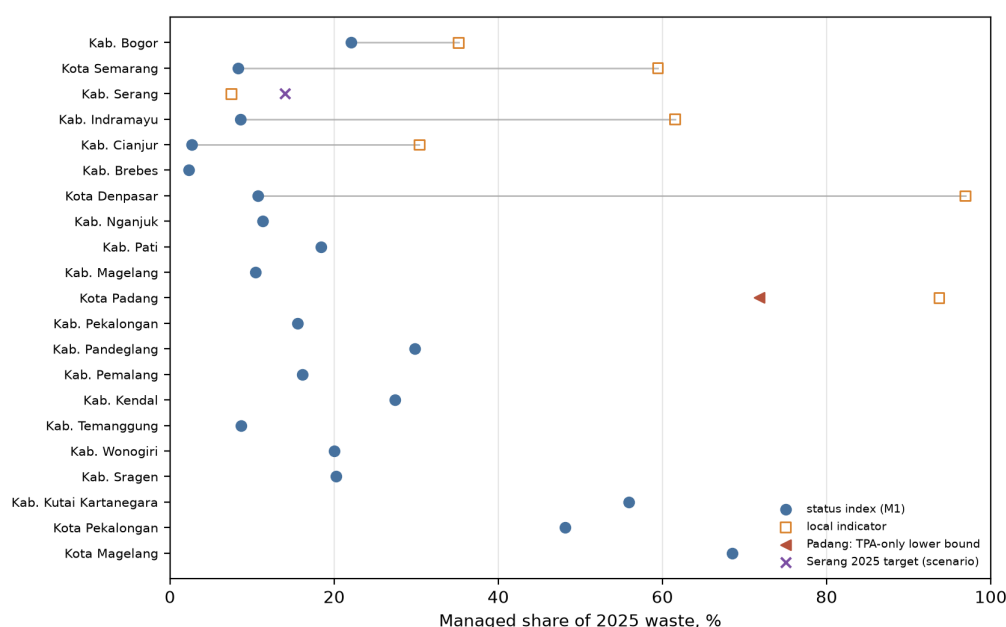


Figure 2: Managed share by indicator family. Grey lines join the two values held for the same location.

3.4 Waste-fraction taxonomy: Level I, II and III

The taxonomy follows the tiered approach of Edjabou et al. (2015): 10 Level I material groups, 36 harmonised Level II fractions (aluminium foil merged into metal packaging, which gives the 36 stated in the source prose while the printed Table 2 lists 37) and a catalogue of 56 Level III refinements. The catalogue is **provisional**: the printed source table has duplicated codes (condoms), WEEE printed under 10.3, index notation for plastics and metals, and no explicit food refinements. The six Level III food sub-fractions are a modelling interpretation. The 56 entries are

neither 56 measurements nor a complete terminal partition.

All joins use the key (level, code), stored as text so that codes such as 6.1 and 6.10 cannot collide. The crosswalk (*taxonomy_crosswalk.csv*) records for every key its parent, the model fraction it maps to and a note where the mapping is not one-to-one. Three cases matter. (i) The ferrous/non-ferrous split (6.x.1, 6.x.2) has the parent *all metal* and crosses 6.1 and 6.2; it is excluded from the terminal partition and kept as a cross-cutting attribute. (ii) The local label of 4.3, *cartons, plates and cups*, is broader than the Danish *beverage cartons*, and beverage cartons appear again as 4.4.1; no Danish value is copied to 4.3. (iii) Level I *Gardening* and *Miscellaneous combustibles* are heterogeneous and do not map to one IPCC category.

Mass balance. The Level III detail file holds only catalogued refinements and covers 76.8 to 93.7% of each stream; it is not forced to 100%. Where a full terminal partition is needed, it is built from the 54 Level III leaves (56 minus the two cross-parent metal rows) and the 24 Level II fractions without children (including 6.1 and 6.2): **78 leaves**, each stream summing to 100%. Because every catalogued parent is fully split by its priors, no residual leaf is needed (the code would create one, labelled *X.R unrefined remainder*, if a gap appeared).

Benchmarks and conditional shares. The Danish single- and multi-family values (Edjabou et al. 2015, Tables 3 and 4) are percentages of total residual household waste. They are a foreign benchmark: not an Indonesian city measurement and not non-domestic waste. A conditional share within the parent is computed only for complete sibling sets on the same basis:

$$p(c | \pi) = \frac{x_c}{\sum_{c' \in \text{ch}(\pi)} x_{c'}} \quad \text{only if every } x_{c'} \text{ is observed} \quad (3)$$

This holds for 11 sibling sets (food, gardening, paper, plastic, metal, glass, miscellaneous combustibles and special waste at Level II; packaging plastic, plastic film and packaging glass at Level III). Board (4.3 missing), inert (9.3, 9.4 missing), garden waste (2.2.4), miscellaneous paper (3.7.4, 3.7.5) and miscellaneous board (4.4.1, 4.4.2) are incomplete and receive no conditional share; partial subsets are not renormalised. Reported zeros are kept; a pseudocount for a Dirichlet prior would be a separate modelling decision.

Allocation priors. The Level I/II/III compositions (*composition_level*_2025.csv*) rescale the package allocations to the 2025 stream masses and label every split with its audit ID. The audit found that several are priors without an Indonesian or literature basis (A01 organic 85/15, A03 lain-lain 80/20, A05 gardening 2/98 where Table 3 gives 9.2/90.8, A09 B3 20/80 where Table 3 implies 28.6/71.4, and the equal-share Level III splits); others are rounded Danish ratios (A04, A06 to A08). These allocations are database content for future calibration. **They are not used by the model**, which computes on the ten model fractions mapped directly from the RIPS categories (Figure 6).

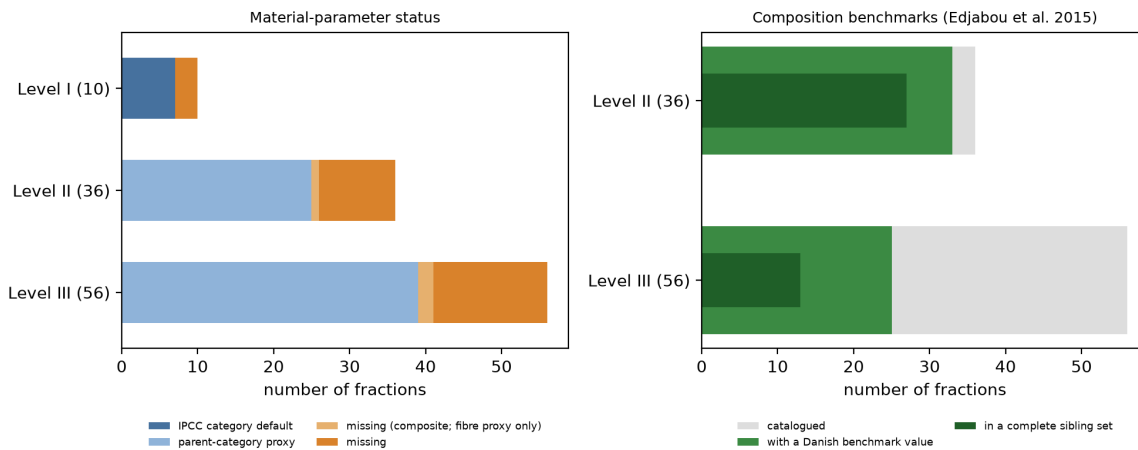


Figure 3: Status of material parameters by level (left) and availability of Danish composition benchmarks (right).

3.5 Material parameters (Table 2 and the detailed table)

IPCC (2006) Volume 5 Table 2.4 supplies default dry matter, DOC, total carbon and fossil-carbon share for the parent categories; the 2019 Refinement Table 3.0 supplies DOCf: 0.70 for food and garden/grass, 0.50 for paper, textile and nappies and 0.10 for wood and branches. DOC is on a dry basis and multiplies dry mass; the wet value is

$$\text{DOC}_{\text{wet}} = \text{DOC}_{\text{dry}} (1 - w) \quad \text{CH}_4 = 10^3 k_D \text{MCF} F \frac{16}{12} \sum_j m_j (1 - w_j) \text{DOC}_{\text{dry},j} \text{DOCf}_j \quad (4)$$

so moisture is applied once. With IPCC moisture the check reproduces the IPCC wet DOC values (food 0.15, garden 0.20, paper 0.40, wood 0.43, textile 0.24, rubber 0.39) within 0.005. Landfill DOCf is not the biochemical methane potential of a digester: AD uses its own yield parameter (A8).

Fraction	w IPCC	w as received (range)	DOC dry	DOCf (status)	C	φ	h dry MJ/kg (status)	τ (status)
food	0.60	0.75 (0.65-0.82)	0.38	0.70 (IPCC 2019)	0.38	0.00	16.5 (legacy)	0.10 (legacy A)
garden	0.60	0.55 (0.40-0.65)	0.49	0.70 (IPCC 2019)	0.49	0.00	16.0 (legacy)	0.40 (legacy A)
paper	0.10	0.35 (0.15-0.50)	0.44	0.50 (IPCC 2019)	0.46	0.01	16.0 (legacy)	0.70 (legacy (Nasrullah))
wood	0.15	0.25 (0.15-0.40)	0.50	0.10 (IPCC 2019)	0.50	0.00	18.0 (legacy)	0.60 (legacy (Nasrullah))
textile	0.20	0.35 (0.15-0.50)	0.30	0.50 (IPCC 2019)	0.50	0.20	18.5 (legacy)	0.70 (legacy A)
rubber	0.16	0.15 (0.05-0.25)	0.47	blank; (gap: scenario 0-0.5)	0.67	0.20	23.0 (legacy)	0.60 (legacy A)
plastic	0.00	0.22 (0.05-0.35)	0.00	0.00 (n/a)	0.75	1.00	30.5 (Gotze 2016, TS)	0.80 (legacy (Nasrullah))
metal	0.00	0.05 (0.02-0.08)	0.00	0.00 (n/a)	0.00	0.00	0.0 (-)	0.00 (-)
glass	0.00	0.02 (0.01-0.03)	0.00	0.00 (n/a)	0.00	0.00	0.0 (-)	0.00 (-)
other	0.10	0.25 (0.10-0.40)	0.00	0.00 (n/a)	0.03	1.00	0.0 (-)	0.10 (legacy A)

Table 2: Properties of the ten model fractions (*table2_model_fraction_parameters.csv*). w, DOC, C and φ: IPCC 2006 Table 2.4 defaults, global, not Indonesia-specific. As-received moisture: analyst assumption carried from v2, calibrated to bulk moisture at Indonesian transfer points (Prabowo et al. 2019). DOCf: IPCC 2019 Table 3.0. h: legacy values cited in v2 to Tchobanoglous et al. (1993), not re-verified, multiplied by k_h ; plastic: aggregate median LHV of 30.5 MJ/kg TS (Götze et al. 2016, Section 3.2.3), not resin-specific and not multiplied by k_h . τ: legacy RDF transfer coefficients; paper, plastic and wood informed by measurements on commercial and industrial waste (Nasrullah et al. 2014), the others analyst assumptions.

Moisture decision. IPCC moisture describes waste as generated; the functional unit is waste as received at the gate, wetter after collection. The main case keeps the v2 as-received values, which bring bulk moisture into the measured Indonesian range, and the IPCC defaults are run as the scenario *moisture_IPCC_default*. Both are internally consistent because DOC, carbon and heating value always multiply the same dry mass. The choice matters (Section 8): it is the single data assumption that changes the most probable pathway most often.

Detailed parameters (*fraction_parameters_L1_L2_L3.csv*, 102 rows). Level II/III values are IPCC parent proxies and keep that status; IPCC did not measure these sub-fractions. Grass, woody material and soil are separated: plant material takes the garden proxy (DOCf 0.70), woody plant material and straw the wood proxy (DOCf 0.10) and humid soil the inert proxy (DOC 0). Composite products receive no whole-product value: coated cartons and cups (4.3, 4.4.1, 4.4.2) keep the paper proxy only for their fibre portion, and composite film, batteries, other HHW, WEEE, tampons and condoms are blank until their material composition is known. Animal-derived food keeps the food proxy with a warning that bones and eggshells lower its degradable carbon. Rubber and leather have no verified DOCf and stay blank. LHV and τ are blank for all sub-fractions. Metal and glass moisture is blank at Level II/III because contamination moisture was not measured; the IPCC dry-matter default of 100% applies at Level I only. Literature from 2026 (the Malang plastic study) is retained as a material benchmark, not as 2025 composition data for the study locations.

Weighted by the terminal composition, a Level II/III DOC value is known for 83 to 96% of each stream's wet mass, moisture and carbon for 78 to 93% and DOCf for 52 to 85% (*outputs/11_parameter_coverage_L2L3.csv*). A Level II model cannot yet be run without imputation, which is why the model stays on the ten model fractions.

4 Model and metamodel

The computation is split into a data layer and a model layer (Figure 5). The data layer turns the package into status-labelled CSV files; the model layer reads them, harmonises, characterises, evaluates the six options, applies the gates and the decision rule, and propagates uncertainty. Figure 6 is the metamodel: the data entities and which of them feed results.

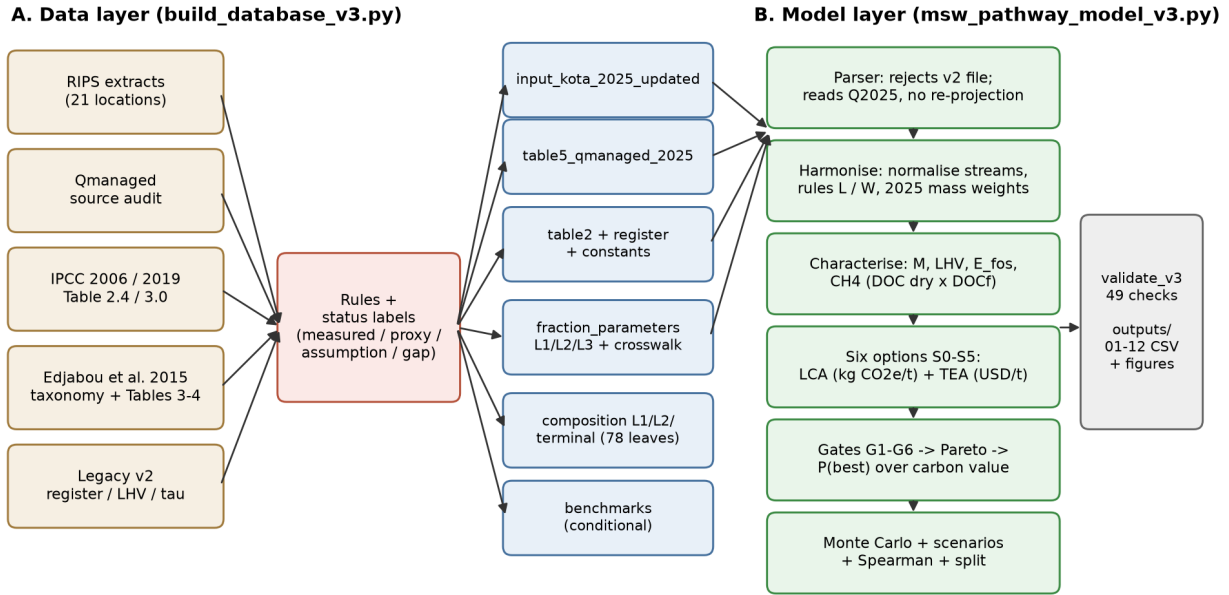


Figure 5: Model flow. Every arrow into the model layer is a CSV file.

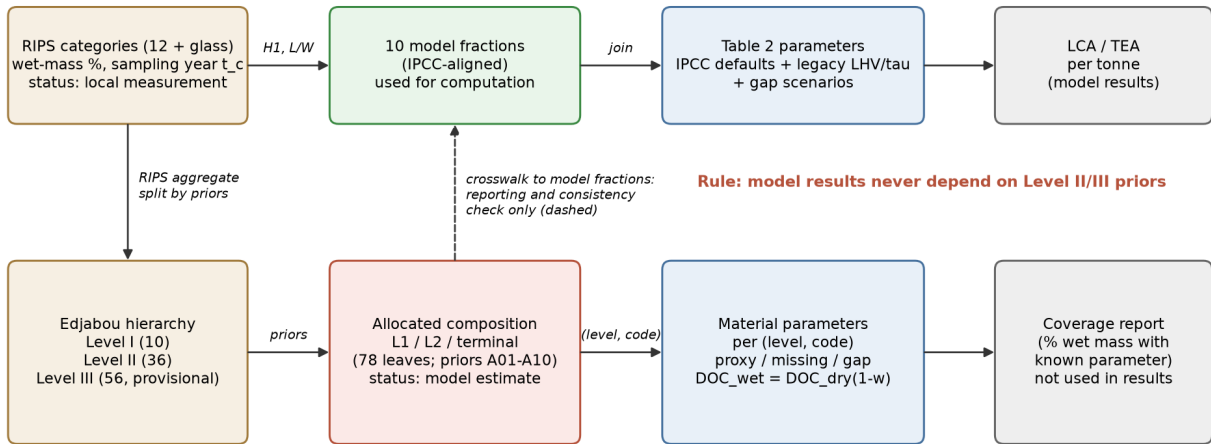


Figure 6: Metamodel. The upper chain produces results; the lower chain (Level I/II/III) is reporting and calibration content and never changes a result.

4.1 Equations

Symbols as in v2: s_j share of fraction j ; w_j moisture; m_j tonnes of fraction j sent to a unit per tonne MSW; h_j dry-matter LHV; C_j dry carbon; ϕ_j fossil share. Characterisation:

$$M = \sum_j s_j w_j \quad H = \max \left[0, \sum_j k_{h,j} s_j h_j (1 - w_j) - \lambda M \right] \quad E_{fos}(m) = 10^3 \frac{44}{12} \sum_j m_j (1 - w_j) C_j \phi_j \quad (5)$$

with $k_{h,j} = k_h$ for legacy heating values and $k_{h,plastic}$ for plastic (register C10, C10b). Landfill (both baselines and every residue landfill):

$$G_{LF}(m, m_{in}) = CH_4(m, MCF) (1 - \eta)(1 - OX) GWP_{CH_4} + a_{LF} \left(\sum_j m_j + m_{in} \right) \quad (6)$$

Pathways (S5 combines the S2 and S3 units; rejects and digestate go to the residue landfill):

$$G_1 = E_{fos}(s) + n_{N_2O} GWP_{N_2O} + a_W - E_{el} EF_{grid} + G_{LF}(0, \alpha_{ash}), \quad E_{el} = \frac{10^3}{3.6} H \eta_W \quad (7)$$

$$G_2 = E_{fos}(r) + e_R EF_{grid} + a_R + M_{del} def_{truck} - \psi E_{net} EF_{coal} + G_{LF}(s - r), \quad r_j = \tau_j k_\tau s_j \quad (8)$$

$$G_3 = V \rho_{CH_4} f_{AD} GWP_{CH_4} + a a_A + \pi (e_R EF_{grid} + a_R) - E_{AD} EF_{grid} + G_{LF}(s - a e_{food}, \delta a) \quad (9)$$

$$G_4 = G_{SL} + P (ef_{PHB} - \sigma ef_{pp}), \quad P = \eta CH_4(s, 1)/R \quad G_5 = G_{AD}(a) + G_{RDF}(s - a e_{food}) + G_{LF}(rej, \delta a) \quad (10)$$

Techno-economic assessment and decision:

$$CRF = \frac{r(1+r)^n}{(1+r)^n - 1}, \quad C_u^{cap} = \frac{CRF K_{ref}(q/0.85/q_{ref})^b}{365 q}, \quad SC_k = C_k + p G_k / 10^3, \quad MAC_k = 10^3 \frac{C_k - C_0}{G_0 - G_k} \quad (11)$$

The gates G1 to G6 (WtE LHV ≥ 7 MJ/kg and $Q \geq 150$ t/d, PSEL tariff at $Q \geq 1,000$ t/d, RDF NCV ≥ 12.56 MJ/kg, kiln within 300 km road distance, PHB ≥ 500 t/yr) are read from *model_constants.csv*. Feasible options are screened by Pareto dominance on (G, C), and the probability of being best is the share of Monte Carlo draws in which an option has the lowest social cost with the carbon value p sampled uniformly from 0 to 100 USD/t CO₂e.

5 Uncertainty: priors, sampling noise and scenarios kept apart

Three kinds of uncertainty are distinguished. (1) **Composition**: a Dirichlet distribution around the RIPS proxy, $S \sim \text{Dir}(\alpha_0 s)$, with $\alpha_0 = 80$, or 40 where the data are flagged. These concentrations are *heuristics*, not statistically calibrated: no replicate sorting campaigns exist for these locations to estimate them. The Dirichlet therefore represents prior uncertainty about how well an 8-day sample of a past year represents 2025, not measured sampling variability. (2) **Parameters**: triangular distributions from the register (Section 6), a common wetness draw for moisture, the harmonisation priors γ and yard, the rubber DOCf gap (0 to 0.5) and the carbon value. (3) **Data scenarios**: discrete alternatives that are not given probabilities (Table S4).

Option	G: composition only	G: parameters only	G: both	C: composition only	C: parameters only	C: both
SL landfill + flare	203	494	543	0	11	11
S1 WtE	210	138	267	12	39	41
S2 RDF	132	310	338	5	15	16
S3 AD	254	386	454	0	15	15
S4 PHB	198	501	547	5	43	43
S5 RDF + AD	110	221	245	5	17	19

Table S3: Median width of the 90% interval over 21 locations when only one source of uncertainty is sampled (G in kg CO₂e/t, C in USD/t; 2,000 draws, market case).

Parameter uncertainty dominates cost; composition contributes a large share of the spread in GHG for WtE (through the plastic share) but less than the parameters for the landfill-based options, where gas collection efficiency dominates.

Scenario	What changes	Status
market	Main case: market prices, no support, residues to sanitary landfill, GWP100, as-received moisture	main
perpres109	WtE electricity at USD 0.20/kWh where $Q \geq 1,000$ t/d	policy
food_separated_at_source	No front-end sorting cost for AD ($\pi = 0$)	what-if
residues_to_open_dump	Residues dumped instead of landfilled (illegal; sensitivity only)	what-if
managed_M1_status_index	Plant size = status-index share $\times Q_{2025}$ (19 locations)	data scenario
managed_M2_local_first	Plant size = local indicator where held, else status index (21; mixed definitions)	data scenario
GWP20	20-year horizon for methane	method
moisture_IPCC_default	IPCC moisture (as generated) instead of as-received values	data scenario
tonnage_overview_projected	Overview totals projected from the composition year	data scenario
docf_rubber_0.5	Rubber/leather DOCf at the IPCC 2006 generic default 0.5	data gap

Table S4: Scenarios. Each uses the same random numbers per location, so differences come only from what the scenario changes. With 4,000 draws, probability differences below about 0.03 are noise.

6 Assumption register

Status: V = checked against the source text during this project; L = literature value recalled from the cited source and still to be checked against the original before submission; A = analyst assumption. Evidence type follows Table 1. The register is *data/assumption_register.csv*; the table below is generated from it.

ID	Key	Meaning	Central (range)	Unit	Source	St.
H11	g_default	Tonnage growth rate where the RIPS gives none (median of six RIPS rates)	0.0173 (0.0065 to 0.0254)	1/yr	RIPS of six locations	A
H12	gamma	Garden share of lumped 'organik' (median and range of 8 locations reporting both)	0.27 (0.04 to 0.62)	-	input_kota; Edjabou et al. 2017	A
H13	yard	Share of 'kayu' treated as garden where kayu >= 10% and no leaf category exists	1 (0.5 to 1)	-	RIPS notes	A
H14	alpha_hi	Dirichlet concentration, unflagged composition (heuristic, not statistically calibrated)	80	-	Analyst heuristic	A
H15	alpha_lo	Dirichlet concentration, flagged composition (heuristic, not statistically calibrated)	40	-	Analyst heuristic	A
C8	F	Methane fraction of landfill gas	0.5 (0.475 to 0.525)	-	IPCC 2006 V5 Ch3	L
C9	doc_k	Multiplier on DOC x DOCf (default uncertainty +-20%)	1 (0.8 to 1.2)	x	IPCC 2006 V5 Ch3 Table 3.5	L
C10	h_k	Calibration factor on legacy dry-matter LHV (not applied to the Gotze plastic value)	0.9 (0.78 to 1.02)	x	Prabowo et al. 2019 (bulk HHV)	V
C10b	h_plastic_k	Uncertainty multiplier on plastic dry LHV (aggregate median, resin mix unknown)	1 (0.9 to 1.1)	x	Analyst range around Gotze et al. 2016 median	A
C11	efg_k	Multiplier on grid emission factor (lower bound = decarbonising grid)	1 (0.7 to 1.1)	x	ESDM factors via JCM 2022	A
C12	docf_rubber	DOCf of rubber/leather: gap scenario (0 = not degradable; 0.5 = IPCC 2006 generic default)	0 (0 to 0.5)	-	IPCC 2006 V5 Ch3 (generic DOCf 0.5); no fraction-specific value	A
B5	mcf_od	Methane correction factor, open dump	0.8 (0.4 to 0.8)	-	IPCC 2006 V5 Ch3 Table 3.1	L
B6	mcf_sl	Methane correction factor, managed anaerobic landfill	1	-	IPCC 2006 V5 Ch3 Table 3.1	L
B7	ox	Oxidation in cover soil, sanitary landfill	0.1 (0 to 0.1)	-	IPCC 2006 V5 Ch3 Table 3.2	L
B8	cap	Lifetime landfill-gas collection efficiency, sanitary landfill	0.5 (0.3 to 0.8)	-	Barlaz et al. 2009; Anshassi et al. 2022; Wei et al. 2024	V
B9	anc_od	Diesel for dump operation	1 (0 to 3)	kg CO2e/t	Manfredi et al. 2009	L
B10	anc_sl	Diesel and materials for sanitary landfill operation	5 (2 to 10)	kg CO2e/t	Manfredi et al. 2009	L
W4	eta_wte	Net electrical efficiency of grate incinerator on low-LHV waste	0.18 (0.14 to 0.22)	-	Astrup et al. 2015; Rand et al. 2000	L
W5	n2o_wte	N2O from continuous stoker incineration	0.05 (0.02 to 0.08)	kg N2O/t	IPCC 2006 V5 Ch5 Table 5.6	L
W6	anc_wte	Auxiliary fuel and flue-gas reagents	10 (5 to 30)	kg CO2e/t	Astrup et al. 2009	L
W7	ash	Bottom ash and APC residue sent to landfill	0.25 (0.15 to 0.3)	t/t	Rand et al. 2000	L
R6	tau_k	Multiplier on RDF transfer coefficients	1 (0.85 to 1.1)	x	Nasrullah et al. 2014	V
R7	omega	Target moisture of RDF (off-taker limit 20-25%)	0.2 (0.15 to 0.25)	-	GIZ ERIC-DKTI 2023	V
R8	q_dry	Heat to evaporate water in the dryer (self-supplied by burning RDF)	3.2 (2.6 to 4)	GJ/t water	Engineering range for rotary dryers	L
R9	e_rdf	Electricity for shredding, screening, baling	40 (20 to 70)	kWh/t	Engineering range for MT plants	A
R10	psi	GJ coal displaced per GJ RDF in the kiln	0.9 (0.8 to 1)	-	Silva et al. 2021; Khandelwal et al. 2019	L
R11	ef_coal	Combustion emission factor of displaced coal (sub-bituminous)	96.1 (94.6 to 101)	kg CO2/GJ	IPCC 2006 V2 Ch2 Table 2.3	L
R12	ef_truck	Truck transport of RDF incl. empty return	0.1 (0.06 to 0.15)	kg CO2e/t-km	GLEC Framework	L
R13	tort	Road distance / straight-line distance to nearest kiln	1.3 (1.2 to 1.5)	x	Analyst assumption	A
R14	anc_rdf	Diesel for loaders at the RDF plant	3 (1 to 6)	kg CO2e/t	Analyst assumption	A
A6	kappa	Share of food waste captured into the digester	0.7 (0.5 to 0.9)	-	Mayer et al. 2019	L
A7	vs_ts	Volatile solids / total solids of food waste	0.87 (0.8 to 0.92)	-	Zhang et al. 2007	L
A8	y_ch4	Methane yield realised in a full-scale digester (BMP-type parameter, not landfill DOCf)	0.36 (0.28 to 0.44)	Nm3 CH4/kg VS	Zhang et al. 2007	L
A9	eta_chp	Electrical efficiency of biogas engine	0.36 (0.32 to 0.4)	-	Mayer et al. 2019	L
A10	par_ad	Own electricity use (pulping, mixing, dewatering)	0.2 (0.1 to 0.3)	-	Mayer et al. 2019	L
A11	fug_ad	Fugitive methane from the biogas plant	0.05 (0.01 to 0.1)	-	IPCC 2006 V5 Ch4	L
A12	anc_ad	Diesel for loaders at the AD plant (per t feed)	3 (1 to 6)	kg CO2e/t	Analyst assumption	A
A13	pre	Front-end separation of food from mixed waste: share of RDF-line cost and electricity	0.4 (0.25 to 0.6)	-	Analyst assumption	A
A14	dig	Dewatered digestate sent to the residue landfill	0.2 (0.1 to 0.35)	t/t feed	Analyst assumption	A
P6	r_phb	Methane per tonne PHB (1.79 as carbon + 0.52 for process energy)	2.3 (2 to 2.8)	t CH4/t PHB	Chidambarampadmavathy et al. 2017	V
P7	ef_phb	Nutrients and extraction chemicals	0.5 (0 to 1.5)	kg CO2e/kg	Rostkowski et al. 2012	L

ID	Key	Meaning	Central (range)	Unit	Source	St.
P8	ef_pp	Cradle-to-gate footprint of displaced polypropylene	1.9 (1.6 to 3.4)	kg CO2e/kg	PlasticsEurope; Chidambarampadmavathy et al. 2017	L
P9	sub_pp	kg PP displaced per kg PHB	1 (0.7 to 1)	kg/kg	Chidambarampadmavathy et al. 2017	L
T7	r	Real discount rate, same for all pathways	0.1 (0.08 to 0.12)	1/yr	Analyst assumption	A
T8	n	Economic lifetime, same for all pathways	20 (15 to 30)	yr	Analyst assumption	A
T9	b	Capacity exponent for CAPEX scaling, same for all units	0.7 (0.6 to 0.85)	-	Tsilemou & Panagiotakopoulos 2006	L
T10	K_wte	WtE CAPEX at reference capacity	120 M (100 M to 200 M)	USD @1000 t/d	Azis et al. 2021 (USD 102.2 M)	A
T11	o_wte	WtE O&M;	27 (21 to 37)	USD/t	Azis et al. 2021; GIZ 2017 in UNDP 2021	V
T12	K_rdf	RDF plant CAPEX at reference capacity (Rp 200 bn)	12.3 M (9 M to 16 M)	USD @300 t/d	GIZ ERIC-DKTI 2023	V
T13	o_rdf	RDF plant O&M;	12 (8 to 21)	USD/t	GIZ ERIC-DKTI 2023; GIZ 2017 in UNDP 2021	V
T14	K_ad	AD plant CAPEX at reference feed capacity	7 M (4 M to 14 M)	USD @100 t/d	GIZ 2017 in UNDP 2021; Aleluia & Ferrao 2017	L
T15	o_ad	AD plant O&M;	14 (10.6 to 30)	USD/t feed	GIZ 2017 in UNDP 2021	V
T16	c_sl	Full cost of sanitary landfill with gas collection and flare	20 (12 to 30)	USD/t @500 t/d	World Bank 2024	V
T17	e_sl	Scale elasticity of landfill cost per tonne	0.3 (0.2 to 0.4)	-	Tsilemou & Panagiotakopoulos 2006	L
T18	c_od	Cost of operating an open dump	4 (2 to 8)	USD/t	Analyst assumption	A
T19	c_phb	PHB production cost excl. methane feedstock	6.8 (4.8 to 9.6)	USD/kg @500 t/a	Listewnik et al. in Chidambarampadmavathy 2017; Levett et al. 2016	A
T20	e_phb	Scale elasticity of PHB unit cost	0.083 (0.05 to 0.15)	-	Levett et al. 2016	A
T21	p_el	Market value of electricity (grid generation cost)	0.07 (0.05 to 0.1)	USD/kWh	Kepmen ESDM 169/2021 (BPP)	L
T22	p_rdf	Price of RDF at kiln gate	1.6 (1 to 2.6)	USD/GJ	Field data; GIZ 2023	A
T23	p_phb	Selling price of PHB	4 (2.5 to 6)	USD/kg	Market reports; Levett et al. 2016	L
T24	c_truck	Cost of trucking RDF	0.1 (0.05 to 0.15)	USD/t-km	Analyst assumption	A

Table 3: Assumption register (61 entries; CED/LU factors in Table 8). New or changed in v3: H14-H15 (heuristic Dirichlet concentrations made explicit), C10b (plastic LHV range), C12 (rubber DOCf gap scenario, previously a fixed 0.5).

7 Validation

validate_v3.py ran 60 checks on the database and the parser; all 60 passed. The groups requested in the brief were all run:

Group	Checks	Examples of what is tested
blank vs zero	6/6	blank DOCf stays blank; parser treats blank and reported-zero glass without imputation; old file rejected
taxonomy	7/7	(level, code) unique in 3 tables; 10/36/56 counts; every parent exists; 6.* documented
mass balance	6/6	M_dom + M_nd = Q2025; Level I = 100%; Level II = parent; terminal partition = 100% (78 leaves)
parent-child	5/5	Level III = Level II parent; detail file partial (76.8-93.7%); conditional shares only for complete sets
projection	6/6	Q2025 = Qt(1+g)^expo once; exponent 0 when already 2025; overview year blank and flagged
Qmanaged	5/5	M1, M2 <= Q2025; Serang 7.45% not 14%; Padang provisional with lower bound
wet/dry	7/7	DOC_dry(1-w) = IPCC wet DOC; LHV dry basis, LHV not HHV; DOCf separate from AD yield; soil/wood separated
provenance	7/7	every row has source, locator and status; proxies keep proxy status; heuristics labelled
CED/LU	7/7	factors have source and status; land take non-negative; landfill area = 1/(rho H) x gross; G and C unchanged
template	4/4	user-template round trip reproduces v3 results; invalid input rejected; blank glass not zero; single projection

Table S5: Validation checks by group (full list in outputs/validation_report.md). In addition the model asserts the single projection for every location, the S5 mass balance and that biogas carbon is below food carbon.

Quantity	Model, 21 locations	Reference
bulk moisture (-)	0.39 to 0.60 (median 0.51)	0.53 to 0.56 at transfer points (Prabowo et al. 2019); 0.64 to 0.66 at a landfill (Fiki et al. 2022)
LHV as received (MJ/kg)	5.0 to 9.5 (median 7.2)	about 5 to 7 (HHV 6.9 to 9.0 measured, Prabowo et al. 2019; 6.5 assumed, Azis et al. 2021)
HHV of dry matter (MJ/kg)	15.3 to 20.3 (median 18.1)	15.7 to 19.2, from HHV 6.9 to 9.0 as received at 53 to 56% moisture (Prabowo et al. 2019)
L0 at MCF = 1 (kg CH4/t)	34 to 65 (median 45)	no Indonesian measurement; reported for transparency
RDF yield (t/t MSW)	0.17 to 0.38 (median 0.27)	0.20 to 0.42 (Indonesian RDF plants, GIZ 2023)
RDF heating value (MJ/kg)	15.0 to 19.8 (median 17.5)	12.6 to 13.8 for biodried whole waste (GIZ 2023); sorted and dried SRF is higher
WtE net electricity (kWh/t)	248 to 474 (median 360)	632 kWh/t at 35% gross conversion (Azis et al. 2021); model uses 18% net

Quantity	Model, 21 locations	Reference
carbon in biogas / carbon in food	0.74	must be below 1

Table 4: Model outputs against published Indonesian values (main case). The model's bulk moisture is at the dry end and its heating value at the high end of the measured range, as in v2.

8 Results for the 21 locations (2025 baseline)

LHV as received is 5.0 to 9.5 MJ/kg (median 7.2). At central values 12 locations pass the WtE heating-value gate G1 and 11 pass G1 and the scale gate G2. 6 locations reach 1,000 t/day on generated waste; on managed waste none does in series M1 and 1 does in series M2 (Kab. Bogor, whose local indicator gives 1,016 t/d).

Option	G	ΔG vs open dump	ΔG vs landfill	C	MAC vs OD	MAC vs SL	Pass
SL landfill + flare	557 (424 to 793)	425 (322 to 609)	-	18 (12 to 40)	34 (21 to 81)	-	21
S1 WtE	121 (-27 to 241)	878 (546 to 1404)	439 (206 to 795)	58 (42 to 120)	62 (36 to 131)	90 (47 to 183)	11
S2 RDF	319 (136 to 421)	661 (455 to 998)	251 (110 to 390)	33 (23 to 61)	43 (31 to 75)	64 (28 to 113)	20
S3 AD	373 (174 to 688)	594 (460 to 775)	173 (104 to 250)	31 (21 to 61)	46 (30 to 97)	69 (40 to 149)	21
S4 PHB	543 (414 to 773)	439 (332 to 628)	14 (10 to 20)	38 (23 to 75)	78 (54 to 154)	1449 (993 to 2412)	20
S5 RDF + AD	142 (13 to 296)	826 (654 to 1106)	420 (309 to 525)	36 (25 to 67)	37 (29 to 70)	42 (25 to 62)	20

Table 6: Pathway results per tonne of MSW, central values, market case: median (minimum to maximum) over the 21 locations, including those failing a gate (last column: number that pass). G, ΔG in kg CO₂e/t; C in USD/t; MAC in USD/t CO₂e.

Location	Flags	SL	S1	S2	S3	S4	S5	0	25	50	100
Kota Padang	N	0.46	0.01	0.01	0.00	0.03	0.49*	SL	SL	S5	S5
Kab. Bogor	G	0.43*	0.00	0.00	0.07	0.09	0.41	SL	SL	S5	S5
Kab. Serang	G	0.54	0.01	0.00	0.06	0.06	0.34	SL	SL	S5	S5
Kota Denpasar	G	0.43	0.16	0.02	0.00	0.06	0.33	SL	SL	S5	S5
Kab. Cianjur	LG	0.48	0.02	0.01	0.02	0.05	0.42	SL	SL	S5	S5
Kab. Indramayu	LG	0.49	0.02	0.01	0.02	0.05	0.42	SL	SL	S5	S5
Kota Semarang	LT	0.47*	0.03	0.01	0.01	0.06	0.42	SL	SL	S5	S5
Kab. Brebes	T	0.46	0.01	0.02	0.00	0.03	0.47*	SL	SL	S5	S5
Kab. Nganjuk	T	0.51	0.03	0.00	0.00	0.04	0.41	SL	SL	S5	S5
Kab. Pati	R	0.29	0.00	0.10	0.00	0.02	0.58	SL	SL	S5	S5
Kab. Magelang	T	0.49*	0.02	0.01	0.00	0.03	0.44	SL	SL	S5	S5
Kab. Pandeanglang		0.58	0.00	0.00	0.08	0.02	0.31	SL	SL	S5	S5
Kab. Pemalang		0.54	0.00	0.01	0.01	0.01	0.43	SL	SL	S5	S5
Kab. Temanggung		0.69	0.01	0.00	0.00	0.02	0.28	SL	SL	SL	S5
Kab. Pekalongan	L	0.39	0.02	0.05	0.00	0.02	0.52	SL	SL	S5	S5
Kab. Sragen	LT	0.49*	0.01	0.04	0.00	0.02	0.44	SL	SL	S5	S5
Kab. Kutai Kartanegara	WT	0.76	0.16	0.00	0.04	0.03	0.00	SL	SL	SL	S1
Kab. Kendal	WT	0.54	0.01	0.00	0.03	0.03	0.38	SL	SL	S5	S5
Kab. Wonogiri	WRT	0.50	0.01	0.01	0.01	0.02	0.44	SL	SL	S5	S5
Kota Pekalongan	L	0.56	0.01	0.04	0.00	0.01	0.38	SL	SL	S5	S5
Kota Magelang	W	0.64	0.00	0.08	0.00	0.00	0.28	SL	SL	SL	S5

Table 7: Probability of being the best feasible option in the market case (carbon value sampled from 0 to 100 USD/t), and the best option at central values for carbon values of 0, 25, 50 and 100. Bold: most probable; *: lead below 0.05. Flags: L lumped organics, W kayu as garden, R rounded percentages, G glass in lain-lain, N glass not reported, T tonnage year unverified.

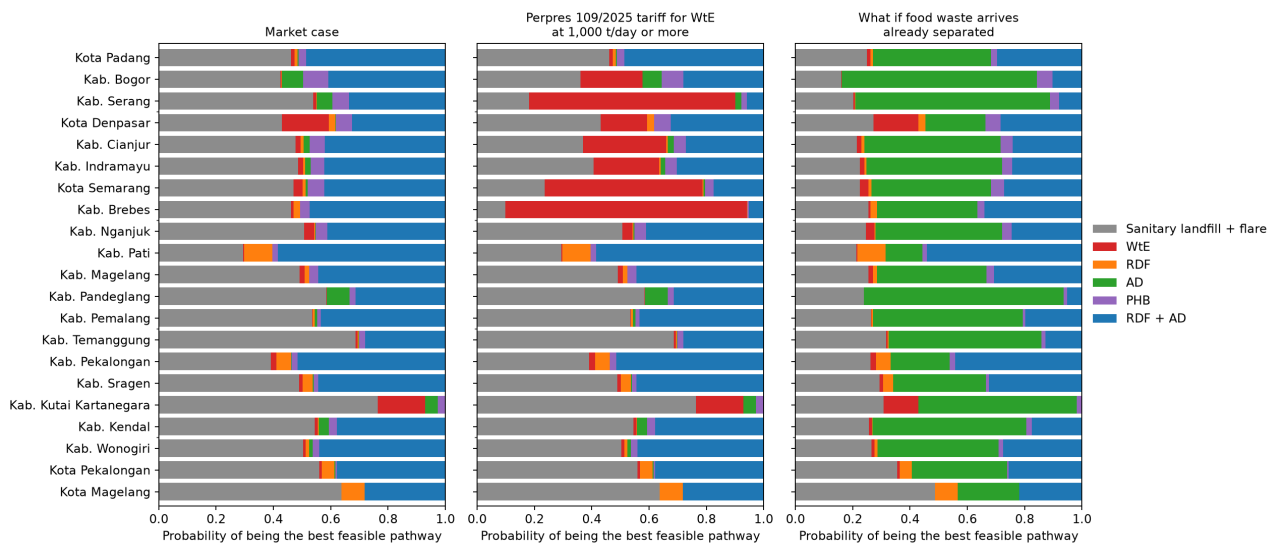


Figure 7: Probability of being the best feasible option in three scenarios.

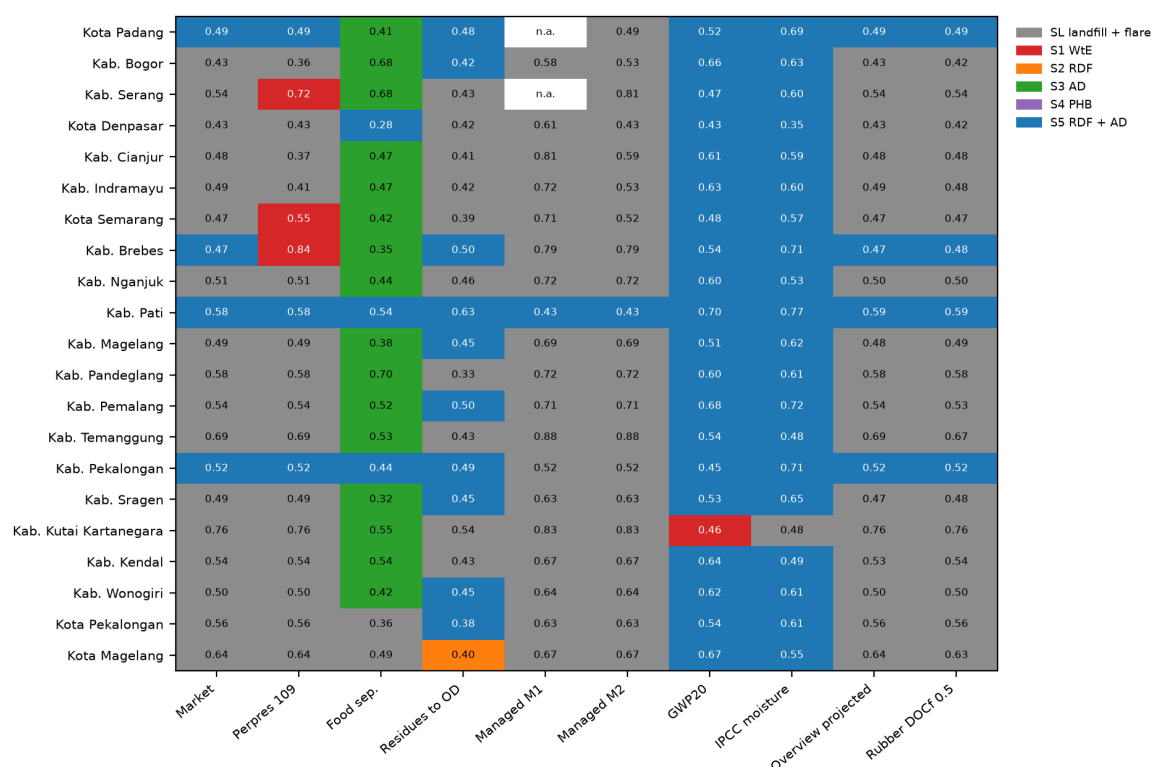


Figure 8: Most probable option per location in each scenario (colour) and its probability (number).

What the numbers say.

- **Landfill first, RDF + AD at moderate carbon values.** Sanitary landfill with flare has the lowest social cost in all 21 locations at 0 and 25 USD/t. RDF + AD is best in 18 locations at USD 50 and 20 at USD 100; its abatement cost against landfill is 42 (25 to 62) USD/t CO₂e. Over the sampled carbon range the two are close: 6 of 21 leads are below 0.05.
- **WtE** needs a heating value of at least 7 MJ/kg, 1,000 t/day and the Perpres 109/2025 tariff together; with the tariff it is most probable in 3 locations (Serang, Semarang, Brebes). Brebes qualifies on generated waste only; 2.3% of it is managed (status index). Semarang's tonnage is an overview total of unverified year.
- **AD alone** becomes most probable in 16 locations if food arrives separated, and almost never for mixed waste. **PHB** is not preferred anywhere; it avoids only 14 (11 to 20) kg CO₂e/t against a landfill at more than USD 990/t CO₂e.
- **Managed tonnage.** On the status-index series (M1) landfill is most probable in 18 of 19 locations; on the local-first series (M2) in 20 of 21. Plants are expensive at today's managed tonnage under both definitions.
- **Data scenarios.** Projecting the overview totals and raising rubber DOCf to 0.5 change no most-probable option. IPCC default moisture changes it in most locations: RDF + AD becomes most probable in 20 locations, because drier waste carries more degradable carbon into the landfill baseline and more heat into RDF. GWP20 has a similar effect (20 RDF + AD, 1 WtE).

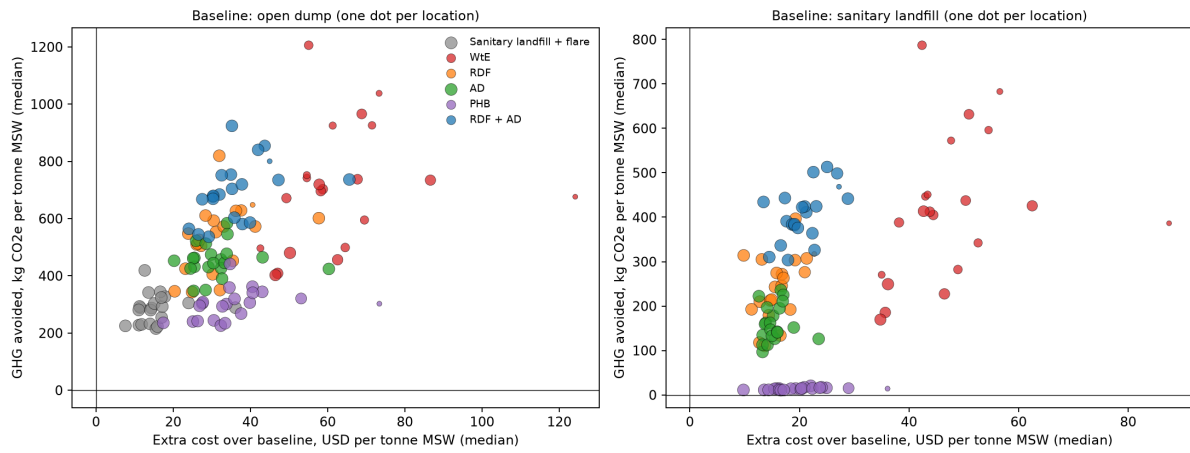


Figure 9: GHG avoided against extra cost, Monte Carlo medians, one dot per location and option.

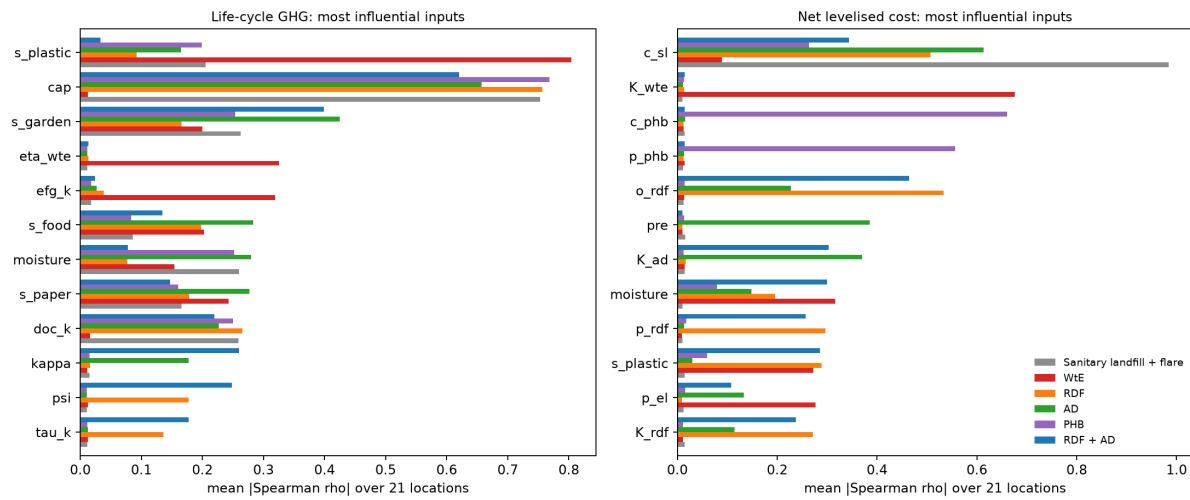


Figure 10: Inputs that drive the results: mean absolute Spearman rank correlation over the locations.

For climate the landfill-gas collection efficiency dominates every option that landfills residues, followed by the plastic share (fossil CO₂ from WtE), the garden share and moisture. For cost: landfill cost, WtE and AD CAPEX, RDF O&M, the AD front-end share and PHB cost and price. These are the values to verify first.

10 Fossil energy (CED) and land take (v3.1)

Two indicators are added to climate change and cost. Both reuse the inventory of the existing model, so they add no new flows, only conversion factors (*data/lcia_factors.csv*, drawn after all v3 parameters so that G and C are unchanged; validation check). **CED** is the non-renewable (fossil) cumulative energy demand in MJ per tonne MSW; biogenic energy is not counted and a negative value is a net saving:

$$CED_k = \sum kWh_{in} PEF + \sum_{ef_{diesel}} \frac{a_i}{\rho_{diesel}} (1 + u_d) - E_{el} PEF - \psi E_{net} (1 + u_c) - P \sigma CED_{pp}, \quad PEF = \frac{3.6}{\eta_{fossil}} (1 + u_f) k_{EF} \quad (12)$$

PEF is the primary fossil energy per kWh of grid electricity, scaled with the same multiplier k_{EF} as the grid emission factor so that a decarbonising grid lowers both. Diesel-type ancillary burdens a_i (kg CO₂e) are converted with the IPCC diesel factor. **Land take** is the landfill area consumed permanently plus plant footprints over their lifetime, in m² per tonne MSW:

$$LU_k = \left(\frac{m_{LF}}{\rho_{SL} H_{SL}} + \frac{m_{inert}}{\rho_{inert} H_{SL}} \right) f_{gross} + \sum_u \frac{fp_u m_u}{0.85 \cdot 365 n}, \quad LU_{OD} = \frac{m}{\rho_{OD} H_{OD}} \quad (13)$$

Key	Meaning	Central (range)	Unit	Source	St.
pef_eta	Net thermal efficiency of the fossil generation displaced or consumed (primary energy = 3.6 / eta MJ per kWh)	0.32 (0.28 to 0.36)	-	PT PLN Indonesia Power fleet thermal efficiency 32.09% in 2022 (annual report, seen as a search-result quotation only); Indramayu CFPP net heat rate about 2,460 kcal/kWh (= 35%) at full load	L
up_fuel	Upstream supply energy of power-plant fuel (mining, processing, transport) as a share of fuel energy	0.08 (0.03 to 0.15)	-	Analyst assumption	A
ef_diesel	Converts diesel-type ancillary burdens (kg CO ₂ e) to diesel energy; the ancillary terms also contain materials, so this is an approximation	0.0741	kg CO ₂ /MJ	IPCC 2006 Vol. 2 Ch. 1 Table 1.4 (gas/diesel oil 74,100 kg CO ₂ /TJ)	L
up_diesel	Upstream (refinery, crude supply) energy of diesel as a share of its energy	0.15 (0.1 to 0.25)	-	Analyst assumption	A
up_coal	Upstream (mining, transport) energy of the kiln coal displaced by RDF	0.05 (0.02 to 0.1)	-	Analyst assumption	A
ef_natgas	Converts the PHB nutrient/chemical burden (kg CO ₂ e) to fossil MJ as if natural-gas based (proxy)	0.0561	kg CO ₂ /MJ	IPCC 2006 Vol. 2 Ch. 1 Table 1.4 (natural gas 56,100 kg CO ₂ /TJ); use as proxy is an assumption	A
ced_pp	Cradle-to-gate fossil CED of displaced polypropylene incl. feedstock energy	75 (65 to 85)	MJ/kg	Order of magnitude of PlasticsEurope PP eco-profile; value not verified in this session	L
rho_sl	In-place density of compacted MSW in a sanitary landfill	0.8 (0.6 to 1)	t/m ³	Engineering range (analyst)	A
h_sl	Effective fill height of a sanitary landfill	20 (10 to 30)	m	Engineering range (analyst)	A
f_gross_sl	Gross site area / fill area (roads, leachate ponds, buffer)	1.3 (1.1 to 1.5)	-	Engineering range (analyst)	A
rho_inert	In-place density of ash, digestate cake and other inert residues	1.3 (1 to 1.6)	t/m ³	Engineering range (analyst)	A
rho_od	Density of uncompacted waste in an open dump	0.5 (0.35 to 0.7)	t/m ³	Engineering range (analyst)	A
h_od	Average heap height of an open dump	5 (3 to 10)	m	Engineering range (analyst)	A
fp_wte	Site footprint of a WtE plant per t/d nameplate capacity	40 (20 to 80)	m ² per t/d	Analyst assumption	A
fp_rdf	Site footprint of an RDF plant (incl. drying and storage) per t/d	50 (25 to 100)	m ² per t/d	Analyst assumption	A
fp_ad	Site footprint of an AD plant (incl. digestate handling) per t/d feed	60 (30 to 120)	m ² per t/d	Analyst assumption	A

Table 8: CED and land-use factors. Most land-use factors are analyst engineering ranges; the thermal efficiency and the polypropylene CED are literature values still to be verified against their originals (status L). An Indonesian TPA design study (Talumelito, Gorontalo) implies about 0.045 m²/t, below the central 0.081 m²/t used here.

Option	CED, GJ/t: median (range)	Saving vs open dump, GJ/t	Land take, m ² /t: median (range)	Saved vs open dump, m ² /t
SL landfill + flare	0.08 (0.08 to 0.08)	-0.06	0.081 (0.081 to 0.081)	0.319
S1 WtE	-4.20 (-5.58 to -2.84)	4.21	0.019 (0.019 to 0.019)	0.381
S2 RDF	-3.80 (-5.77 to -1.81)	3.82	0.061 (0.053 to 0.066)	0.339
S3 AD	-0.47 (-0.77 to -0.19)	0.48	0.067 (0.059 to 0.073)	0.333
S4 PHB	-0.57 (-0.85 to -0.42)	0.59	0.081 (0.081 to 0.081)	0.319
S5 RDF + AD	-4.49 (-6.29 to -2.78)	4.50	0.045 (0.042 to 0.051)	0.355

Table 9: Fossil CED and land take per tonne MSW, central values, market case, 21 locations.

WtE, RDF and RDF + AD save 4.2, 3.8 and 4.5 GJ of fossil energy per tonne (displaced grid electricity and kiln coal), AD alone 0.5 GJ and PHB 0.6 GJ (displaced polypropylene); the sanitary landfill consumes 78 MJ for operation. Land take is dominated by landfilling: an open dump uses about 0.40 m²/t, a sanitary landfill 0.081, RDF + AD 0.045 and WtE 0.019 m²/t, mostly for ash. The energy ranking follows the climate ranking; land take adds a reason to divert waste from landfill where land is scarce, such as Java.

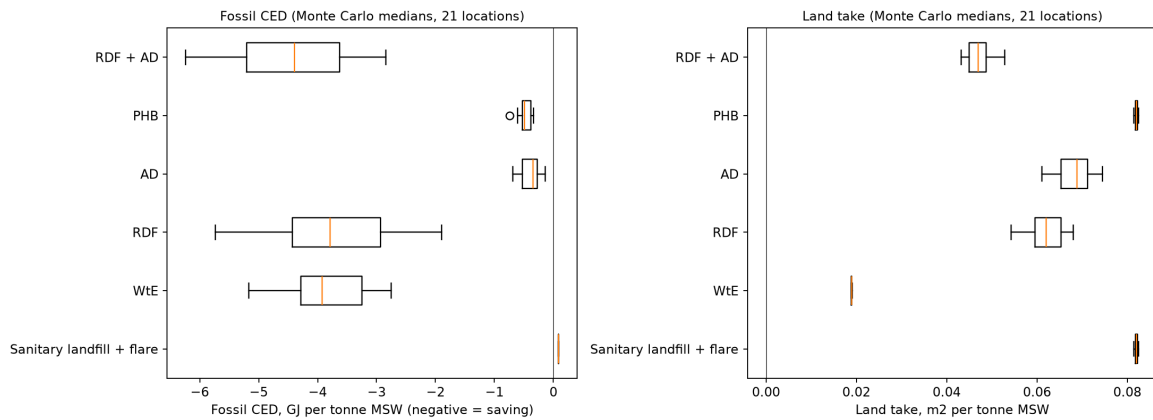


Figure 11: Fossil CED and land take of each option, Monte Carlo medians over the 21 locations.

11 Conditions under which each option becomes preferable: scenario discovery

The research question asks for conditions, not for a ranking of 21 cases. The model was therefore run over 40,000 combinations of conditions that a planner can observe or choose: carbon value (0 to 100 USD/t CO₂e), waste to the facility (50 to 3,000 t/day, log-uniform), road distance to a cement kiln (10 to 400 km), grid region, availability of the Perpres 109/2025 tariff, source separation of food, composition, moisture, landfill-gas collection and landfill cost; all other parameters were sampled from the register. Composition was drawn from a Dirichlet distribution centred on the mean of the 21 RIPS compositions with concentration 46 fitted to their between-city spread (method of moments), so the rules describe compositions like those observed and are extrapolations outside them. For each draw the best feasible option (lowest social cost) was recorded. A classification tree of depth 4 reproduces that choice in 70% of held-out draws (majority-class baseline 43%); PRIM (Friedman and Fisher 1999) searched for boxes of conditions in which one option is best with high density.

Rule (conditions)	Best option	Purity	Share of draws
food separated = yes and carbon value > 53.2 and Q tpd <= 1e+03	S3 AD	0.69	17%
food separated = no and carbon value <= 39.6 and Q tpd <= 1e+03	SL landfill + flare	0.95	15%
food separated = no and carbon value > 55.6 and kiln km <= 299 and Q tpd <= 1.02e+03	S5 RDF + AD	0.64	12%
food separated = yes and 29.2 < carbon value <= 53.2 and Q tpd <= 1e+03	S3 AD	0.63	8%
food separated = no and 39.6 < carbon value <= 55.6 and Q tpd <= 1e+03	SL landfill + flare	0.75	6%
food separated = yes and 13.5 < carbon value <= 29.2 and Q tpd <= 979	SL landfill + flare	0.63	6%
food separated = yes and carbon value > 29.2 and Q tpd > 1e+03 and LHV > 7	S1 WtE	0.52	5%
food separated = yes and carbon value <= 13.5 and Q tpd <= 933	SL landfill + flare	0.91	5%
food separated = no and carbon value > 55.6 and kiln km <= 299 and Q tpd > 1.02e+03	S5 RDF + AD	0.52	4%
food separated = yes and carbon value > 29.2 and Q tpd > 1e+03 and LHV <= 7	S3 AD	0.75	4%
food separated = no and carbon value <= 55.6 and Q tpd > 1e+03 and tariff = no	SL landfill + flare	0.72	4%
food separated = no and carbon value <= 55.6 and Q tpd > 1e+03 and tariff = yes	S1 WtE	0.55	3%

Table 10: Leaves of the decision tree covering at least 3% of draws (outputs/discovery_cart_rules.csv). Purity: share of the draws in the leaf where the predicted option is indeed best. Q tpd: t/day to the facility; kiln km: road distance.

Option	Base rate	Coverage	Density	Box (conditions)
SL landfill + flare	0.435	0.31	0.80	carbon_value: 0.00119 to 65.7; tariff = 0; food_separated = 0
S1 WtE	0.085	0.42	0.97	Q_tpd: 1.01e+03 to 3e+03; tariff = 1; food_separated = 0; LHV: 6.98 to 14.6
S2 RDF	0.006	n.a.	n.a.	best in fewer than 3% of draws; no box
S3 AD	0.276	0.31	0.74	carbon_value: 33.1 to 100; kiln_km: 29.9 to 400; tariff = 0; food_separated = 1; food_share: 0.38 to 0.736; paper_garden_share: 0.0311 to 0.301; moisture: 0.409 to 0.68; LHV: 2.58 to 10
S4 PHB	0.019	n.a.	n.a.	best in fewer than 3% of draws; no box
S5 RDF + AD	0.180	0.31	0.68	carbon_value: 50.6 to 100; kiln_km: 10 to 296; tariff = 0; food_separated = 0; gas_capture: 0.303 to 0.68

Table 11: PRIM boxes. Coverage: share of all draws where the option is best that fall in the box; density: share of draws in the box where it is best. RDF alone and PHB are best in fewer than 3% of draws.

Reading the rules. The carbon value (importance 0.41), source separation of food (0.31), plant scale (0.12) and kiln distance (0.08) carry the decision; heating value and the tariff matter only for WtE. For **mixed waste without the WtE tariff**, the sanitary landfill with flare is best in 62% of draws and RDF + AD in 27%. The rules are:

- **Sanitary landfill with flare** is preferred when the carbon value is below about 40 USD/t (purity 0.95), and above that when no cement kiln lies within about 300 km.
- **RDF + AD** is preferred for mixed waste when the carbon value exceeds about 55 USD/t and a kiln lies within about 300 km, at any scale (PRIM: carbon value above 51 USD/t, kiln within 296 km, gas collection below 0.68).
- **WtE** is preferred when the Perpres 109/2025 tariff applies, at least 1,000 t/day are delivered and the LHV is at least 7 MJ/kg (PRIM density 0.97); without the tariff it is rarely best.
- **AD alone** is preferred when food waste arrives separated at source and the carbon value exceeds about 30 USD/t, below 1,000 t/day.
- **RDF alone and PHB** are best in fewer than 3% of all draws: they are dominated by RDF + AD and by flaring.

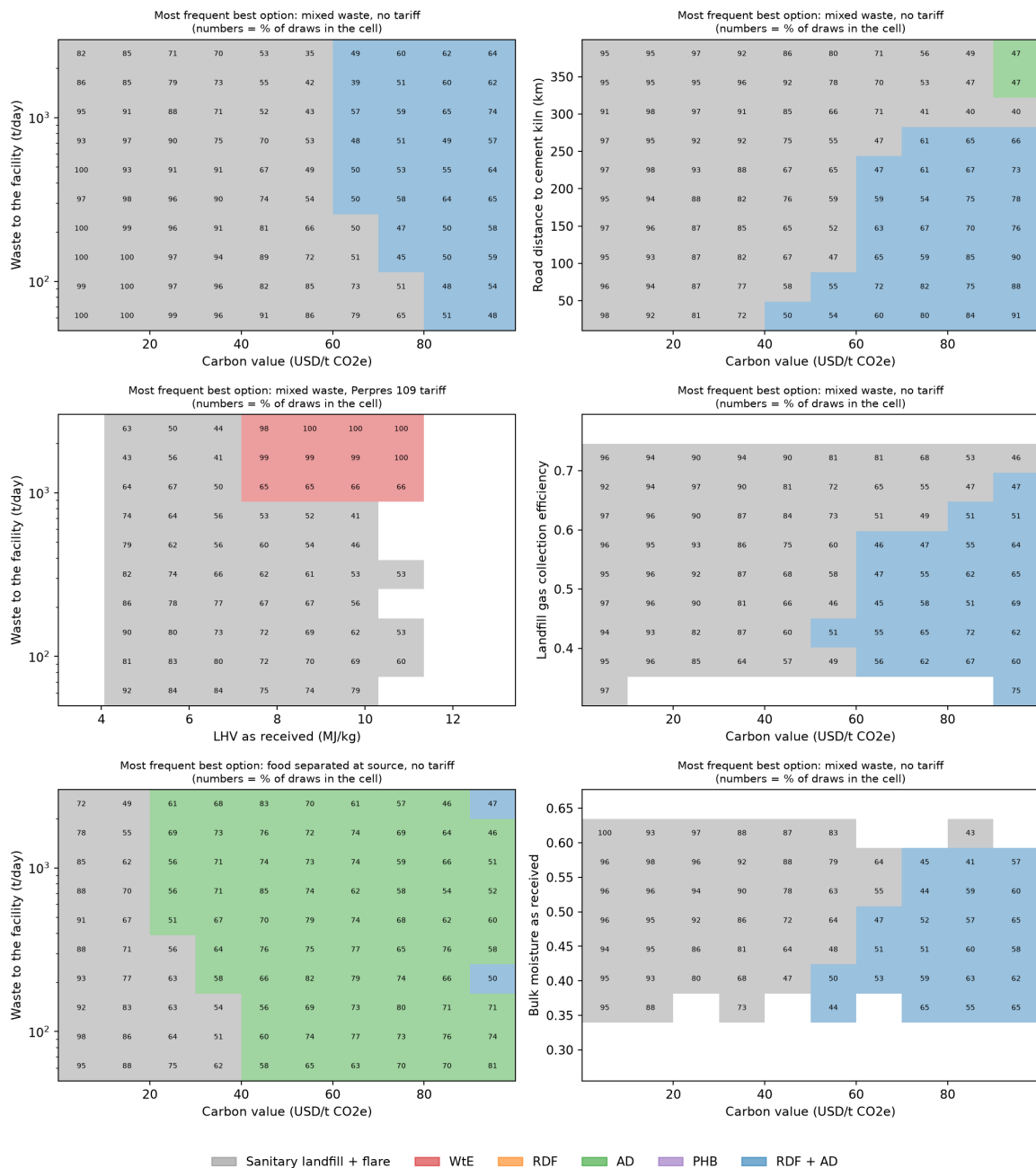


Figure 12: Condition maps. Each cell shows the most frequent best option and the share of draws in which it is best, separately for mixed waste without tariff, mixed waste with the Perpres 109/2025 tariff and food separated at source.

Sobol indices. To complement the rank correlations, first-order and total Sobol indices (Saltelli sampling, $N = 512$ base samples, 58 register inputs plus the common wetness draw; composition fixed at each location's value) were computed for every location and averaged.

Output	Most influential inputs: total index ST (first-order S1)
Social-cost gap RDF + AD minus landfill at 50 USD/t	moisture (wetness) 0.28 (0.26); landfill-gas collection 0.23 (0.22); RDF O&M 0.10 (0.09); AD CAPEX 0.06 (0.04); RDF price 0.04 (0.04); landfill cost 0.04 (0.03)
GHG of the sanitary landfill	landfill-gas collection 0.68 (0.65); moisture (wetness) 0.20 (0.19); DOC × DOCf multiplier 0.09 (0.08); garden share of organik 0.03 (0.03)
GHG of WtE	grid factor multiplier 0.43 (0.42); WtE efficiency 0.41 (0.41); moisture (wetness) 0.08 (0.06); LHV calibration 0.05 (0.05)

Table 12: Sobol indices, mean over the 21 locations (outputs/discovery_sobol_mean.csv).

Moisture as received and landfill-gas collection efficiency together explain about half of the variance of the social-cost gap between RDF + AD and the landfill, which is why the moisture scenario of Section 8 can reverse the ranking. These two quantities, together with RDF operating cost and price, are the data to measure first.

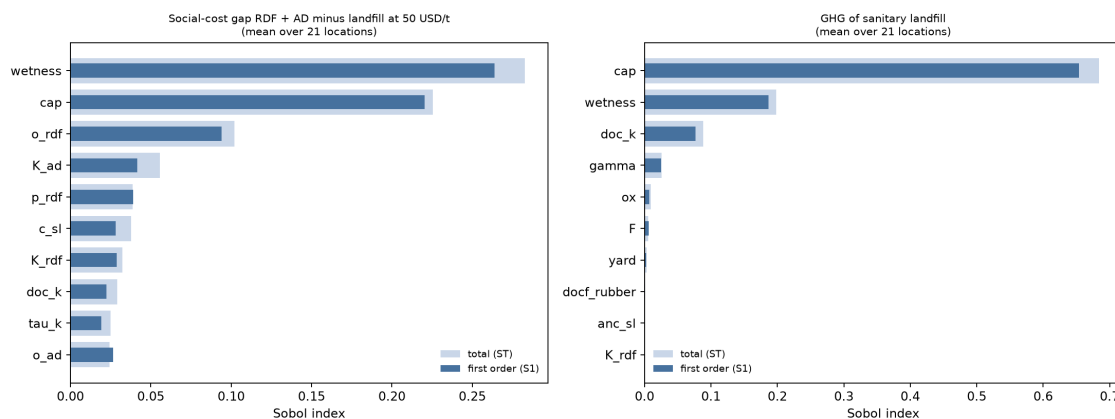


Figure 13: Sobol indices for the social-cost gap RDF + AD minus landfill and for the GHG of the landfill, mean over 21 locations.

12 A reproducible decision tool for researchers and planners

The model is packaged as the Python package *mswpath* (core model, input template and validator, scenario discovery, reports and an interactive form) and exposed through *MSW_Decision_Tool_Colab.ipynb*, which runs in Google Colab without installation. The notebook (i) reproduces the results for the 21 RIPS locations; (ii) accepts a one-row-per-city template (*templates/city_input_template.xlsx*, with an English/Indonesian column guide), validates it, applies the same single-projection and normalisation rules and reports every assumption it applies; (iii) offers an interactive form for one city; (iv) re-runs scenario discovery and applies the decision tree to a city described by the user; and (v) writes Excel and HTML reports in English or Indonesian that carry the caveats and the data-quality warnings. A round-trip test confirms that two RIPS locations entered through the template reproduce the database results. Versions are pinned in *requirements.txt*, random seeds are fixed and *CITATION.cff* gives the citation.

13 Limits of validity

- **Temporal proxy.** Compositions sampled 2014 to 2025 represent 2025 without a trend; eight tonnages have an unverified year. Results describe a 2025 baseline *conditional* on these proxies, not a measured 2025 state.
- **Managed-waste definitions.** No single indicator exists for all 21 locations. Rankings on managed tonnage are valid only within series M1, and series M2 mixes definitions. Padang's share is provisional.
- **Moisture and heating value.** No fraction-level moisture or LHV has been measured for the study locations; the as-received moisture is calibrated to two published samples and the LHV values are legacy. The moisture scenario shows that this choice can reverse the ranking between landfill and RDF + AD. Local proximate analysis is the priority.
- **Level II/III.** Level II/III compositions are model allocations and their parameters are proxies; they cannot support fraction-level conclusions (for example on recycling of specific resins) until local sorting data exist.
- **Foreign benchmarks.** Danish residual household waste is not Indonesian mixed MSW and not non-domestic waste.

- **Global defaults.** IPCC defaults are global and describe waste as generated, not at the landfill gate.
- **Screening level.** One impact category; no time dynamics of landfill gas; class-5 costs not escalated; provisional coordinates; independent triangular inputs; Status L values still to be checked against the originals.
- **Coverage.** 21 locations, 12 in Central Java; conclusions about Indonesian conditions in general need more regions.

14 Data gaps carried forward

Measured fraction-level moisture and LHV (local proximate analysis); DOCf for rubber and leather; material composition of coated board, composite film, batteries, HHW and WEEE; Level III food and miscellaneous-paper sorting data; the year of the overview tonnage index; the definition and denominator of the RIPS status index; whether Padang's 141.76 t/d is actual or potential recovery; chlorine and sulphur of RDF; verified plant and kiln coordinates; the numerical tables of Riber et al. (2009). The full list with the decision taken for each item is in CHANGELOG_AND_SOURCE_DECISIONS.md.

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